

EXPLORATORY DATA ANALYSIS (EDA) REPORT

House Price Prediction Using Machine Learning

Dataset: Ames Housing Dataset

Project Type: Regression Analysis

Tools & Technologies: Python, Pandas, NumPy, Matplotlib, Seaborn

1. Introduction

1.1 Overview

Exploratory Data Analysis (EDA) is an important step in the machine learning workflow. It helps understand the dataset by analyzing feature distributions, identifying relationships between variables, detecting outliers, and discovering patterns that influence the target variable.

In this project, EDA was performed to gain insights into the Ames Housing dataset before model development.

2. Objectives

The objectives of Exploratory Data Analysis are:

- To understand the distribution of housing features.
- To analyze the target variable (SalePrice).
- To identify highly correlated features.
- To detect outliers.
- To measure feature skewness.
- To understand multicollinearity among features.

3. Statistical Summary

A descriptive statistical analysis was performed to understand the central tendency and spread of numerical features.

The analysis included:

- Mean
- Median
- Standard Deviation
- Minimum Value
- Maximum Value
- Quartiles (Q1, Q2, Q3)

These statistics provide a general overview of the dataset.

	Id	MSSubClass	LotFrontage	LotArea	OverallQual	OverallCond	YearBuilt	YearRemodAdd
count	1460.000000	1460.000000	1201.000000	1460.000000	1460.000000	1460.000000	1460.000000	1460.000000
mean	730.500000	56.897260	70.049958	10516.828082	6.099315	5.575342	1971.267808	1984.865753
std	421.610009	42.300571	24.284752	9981.264932	1.382997	1.112799	30.202904	20.645407
min	1.000000	20.000000	21.000000	1300.000000	1.000000	1.000000	1872.000000	1950.000000
25%	365.750000	20.000000	59.000000	7553.500000	5.000000	5.000000	1954.000000	1967.000000
50%	730.500000	50.000000	69.000000	9478.500000	6.000000	5.000000	1973.000000	1994.000000
75%	1095.250000	70.000000	80.000000	11601.500000	7.000000	6.000000	2000.000000	2004.000000
max	1460.000000	190.000000	313.000000	215245.000000	10.000000	9.000000	2010.000000	2010.000000

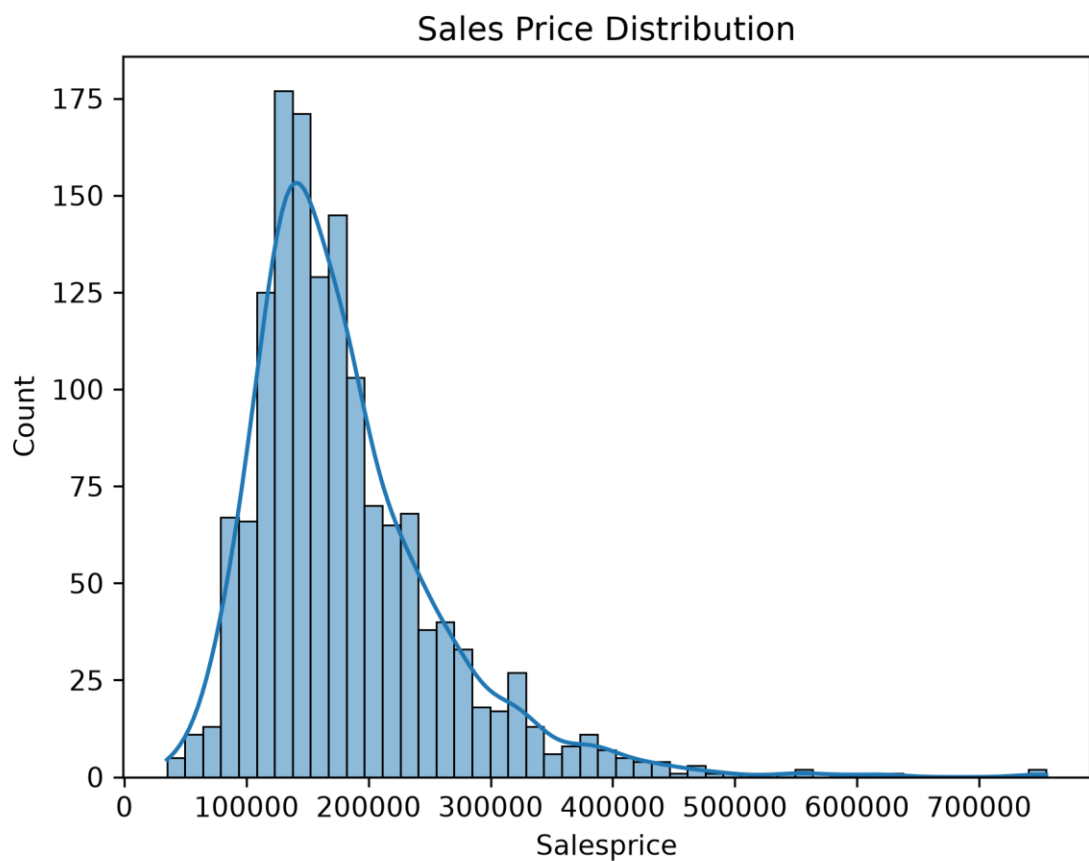
4. Target Variable Analysis

The target variable **SalePrice** was analyzed to understand its distribution.

Initially, the distribution showed positive skewness, indicating that most houses were sold within a moderate price range while a smaller number of houses had significantly higher prices.

Key Observation

- Distribution was right-skewed.
- Few expensive houses created a long tail.
- Transformation techniques were considered during preprocessing.



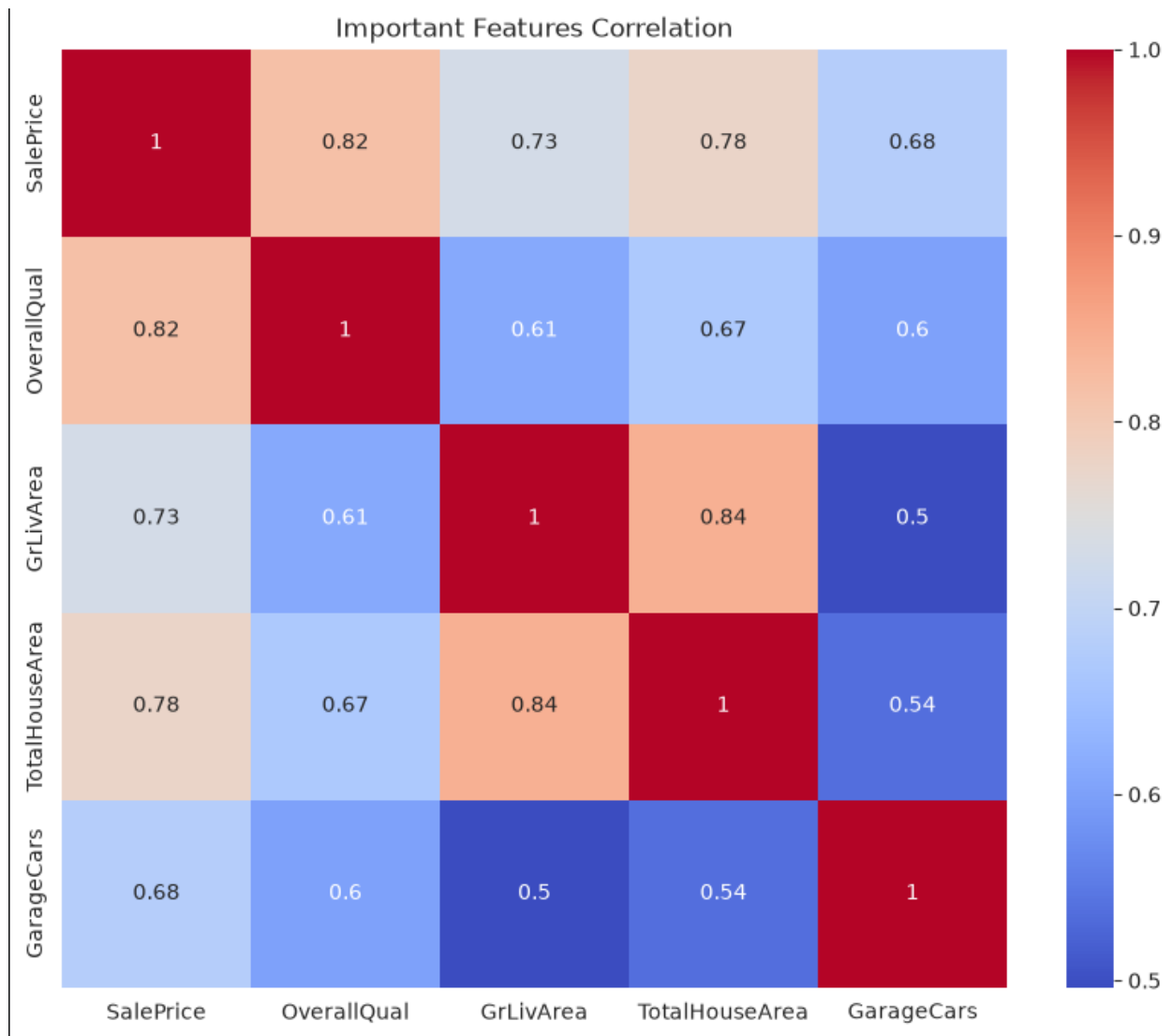
5. Correlation Analysis

Correlation analysis was performed to identify features that strongly influence house prices.

The correlation matrix revealed that several variables had a strong positive relationship with the target variable.

Top Correlated Features

Feature	Correlation with SalePrice
OverallQual	0.82
TotalHouseArea	0.78
GrLivArea	0.73
GarageCars	0.68
TotalBathrooms	0.64



6. Outlier Analysis

Outliers were detected using the **Interquartile Range (IQR)** method.

The analysis identified observations with extremely high or low house prices.

IQR Formula

- Q1 = 25th Percentile
- Q3 = 75th Percentile

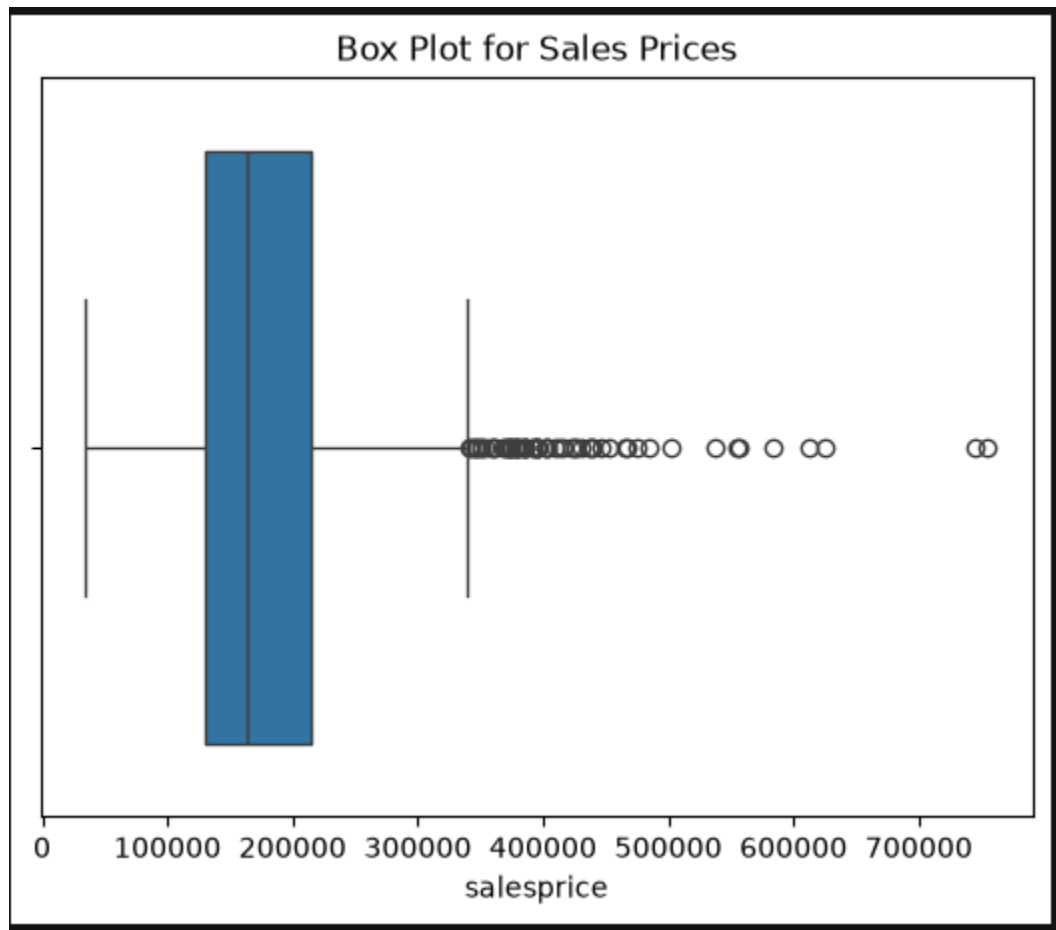
- $IQR = Q3 - Q1$

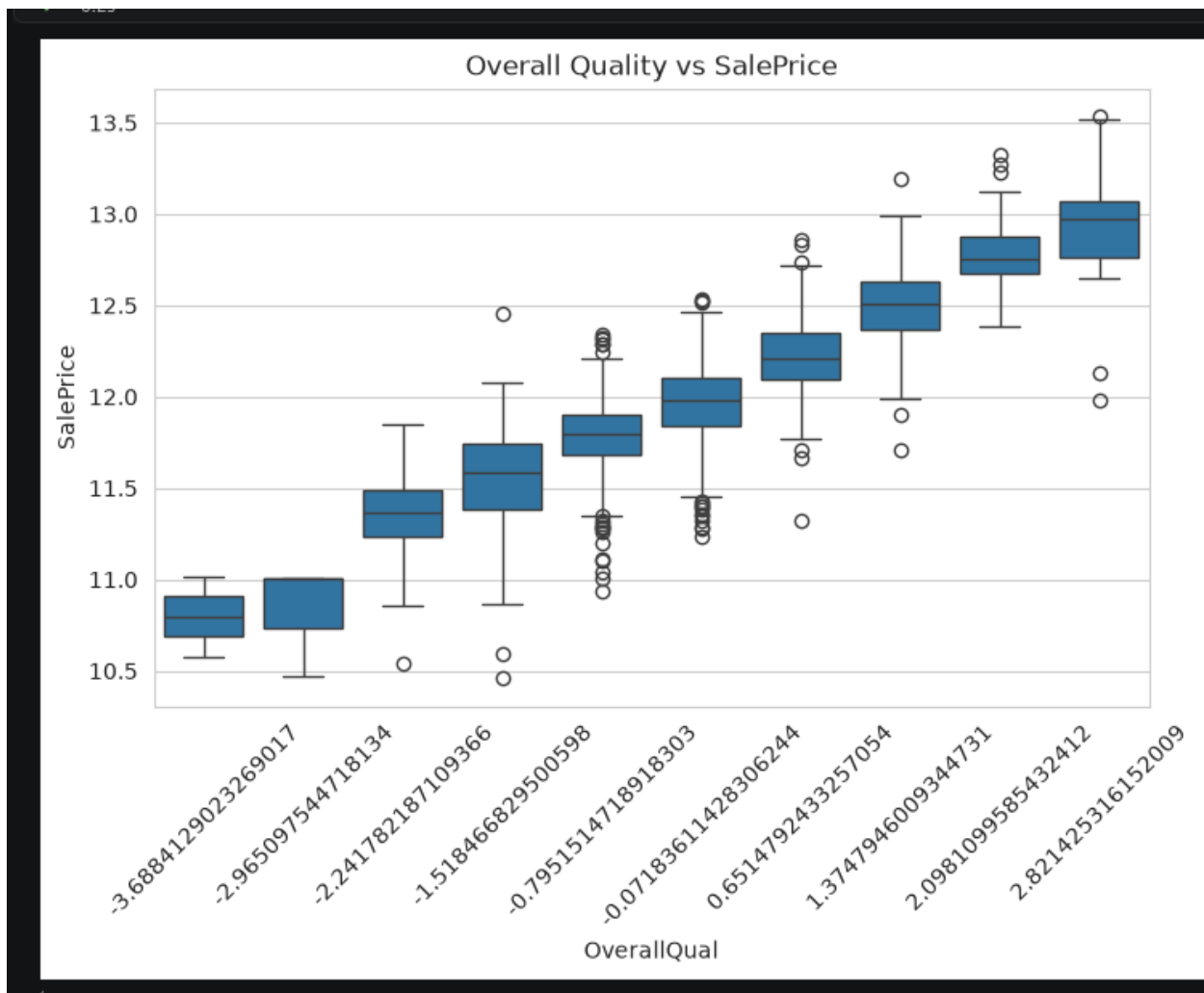
Lower Bound = $Q1 - 1.5 \times IQR$

Upper Bound = $Q3 + 1.5 \times IQR$

Observation

The detected outliers were analyzed before model development to understand their impact on prediction performance.





7. Skewness Analysis

Skewness measures the asymmetry of a feature's distribution.

Features with high skewness were identified because they can negatively affect machine learning models.

Skewness Interpretation

Skewness Value	Interpretation
-1 to 1	Approximately Normal
> 1	Highly Positively Skewed
< -1	Highly Negatively Skewed

MiscVal	24.476794
PoolArea	14.828374
LotArea	12.207688
3SsnPorch	10.304342
LowQualFinSF	9.011341
KitchenAbvGr	4.488397
BsmtFinSF2	4.255261
ScreenPorch	4.122214
BsmtHalfBath	4.103403
EnclosedPorch	3.089872
MasVnrArea	2.677616
LotFrontage	2.409147
OpenPorchSF	2.364342
SalePrice	1.882876
BsmtFinSF1	1.685503
WoodDeckSF	1.541376
TotalBsmtSF	1.524255
MSSubClass	1.407657
1stFlrSF	1.376757
GrLivArea	1.366560
BsmtUnfSF	0.920268
2ndFlrSF	0.813030
OverallCond	0.693067
TotRmsAbvGrd	0.676341
HalfBath	0.675897
...	
YearRemodAdd	-0.503562
YearBuilt	-0.613461
GarageYrBlt	-0.678333

8. Multicollinearity Analysis

Multicollinearity occurs when independent variables are highly correlated with each other.

Variance Inflation Factor (VIF) analysis was performed to detect multicollinearity.

VIF Interpretation

VIF Value	Interpretation
1–5	Low Multicollinearity
5–10	Moderate Multicollinearity
>10	High Multicollinearity

	Feature	VIF
219	PoolQC_None	27975.551254
224	MiscFeature_None	15145.973819
103	RoofMatl_CompShg	7995.482136
214	GarageCond_TA	2509.574037
210	GarageQual_TA	2211.932252
172	Heating_GasA	1738.158063
81	Condition2_Norm	1140.291558
23	TotalHouseArea	1008.580764
145	ExterCond_TA	729.141159
98	RoofStyle_Gable	726.633496
226	MiscFeature_Shed	485.052214
32	Street_Pave	406.415486
30	MSZoning_RL	230.827481
100	RoofStyle_Hip	183.693191
120	Exterior1st_VinylSd	173.309852
19	PoolArea	163.472658
134	Exterior2nd_VinylSd	160.962335
200	GarageType_Attchd	147.811857
194	Functional_Typ	144.231984

9. Key Insights

The exploratory analysis revealed several important findings:

- SalePrice showed a positively skewed distribution.
- Overall Quality and Total House Area had the strongest relationship with SalePrice.
- Some features contained significant outliers.
- Feature engineering improved feature representation.
- Multicollinearity was analyzed using VIF.
- Skewness analysis identified variables requiring transformation.

10. Conclusion

Exploratory Data Analysis provided valuable insights into the structure and characteristics of the Ames Housing dataset. The analysis helped identify important features, detect outliers, evaluate skewness, and understand relationships between variables.

These findings guided the preprocessing and feature engineering steps, ensuring that the dataset was well-prepared for training machine learning regression models.